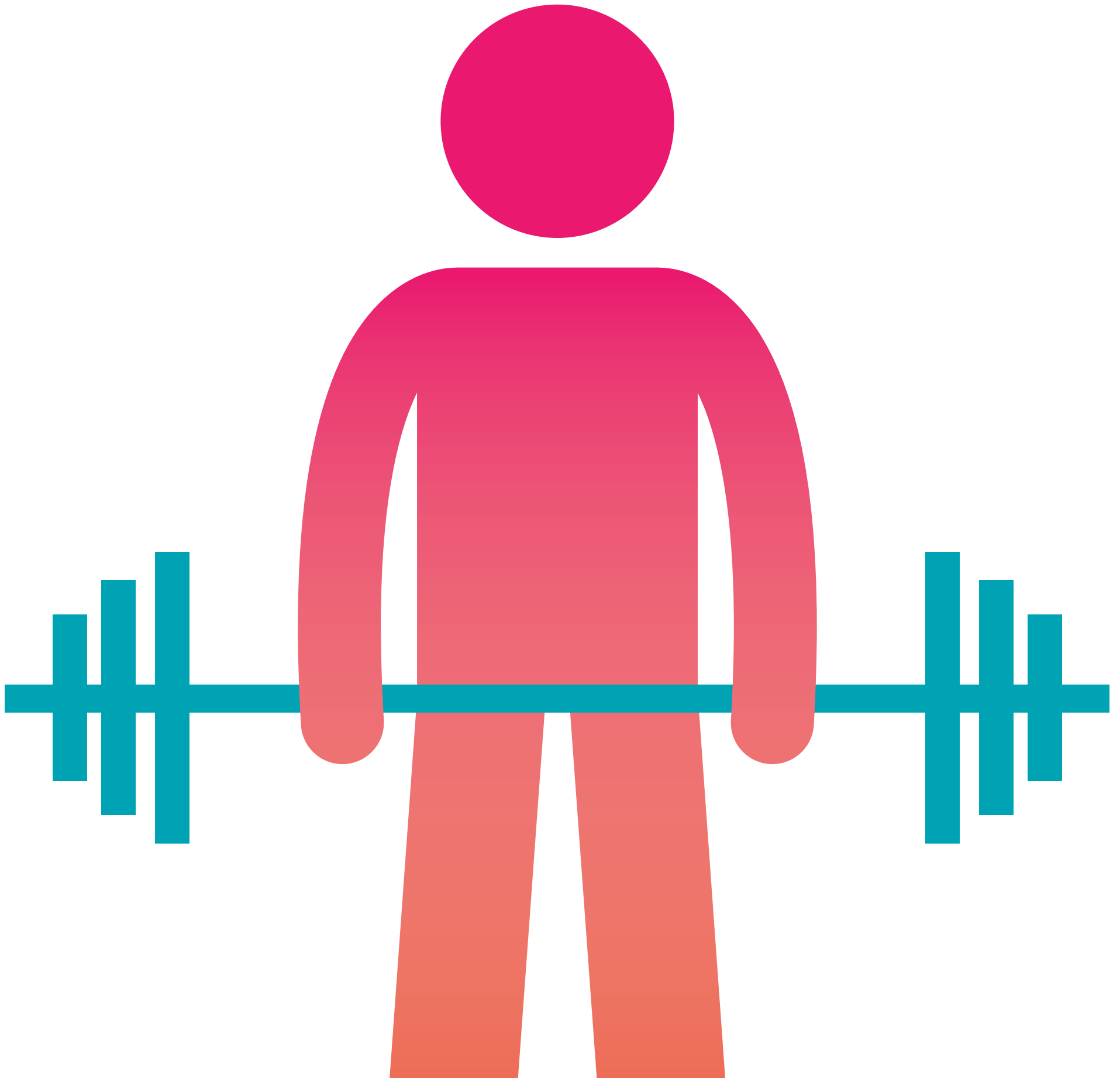


Examine.com

Muscle Gain & Exercise Performance Stack Guide



Written by the editors of Examine.com
Updated November 21, 2016

Medical Disclaimer

This guide is a general health-related information product intended for adults over the age of 18.

This guide is for educational purposes only. It does not constitute medical advice. Please consult a medical or health professional before you begin any exercise, nutrition, or supplementation program, or if you have questions about your health.

If you choose to engage in any activity or use any product mentioned in this guide, you do so of your own free will, and you knowingly and voluntarily accept the risks.

While we mention major known interactions, it is possible for any supplement to interact with other supplements, as well as with foods and pharmaceuticals. Therefore, it is important to consult a medical professional prior to using any supplement mentioned in this guide.

Specific study results mentioned in this guide should not be considered representative of typical results. Individual results do vary.

Keep in mind that not all supplements contain the exact compounds and amounts listed on the label. Always investigate supplement companies, as well as the supplement itself, before purchasing anything. Also note that, more than isolated compounds, herbs can have variability from batch to batch, which can alter their efficacy and safety.

For evidence supporting the claims mentioned in this guide, please visit [Examine.com](https://www.examine.com).

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How to Use This Guide

The team at Examine.com has been publishing research on nutrition and supplementation since March 2011. Drawing from all we've learned, we designed this Stack Guide to help you figure out which supplements can help you reach your health goal, and which can hinder you or just waste your money.

Core supplements are the supplements most likely to help, while having little to no side effects. They tend to have more research backing than do the other supplements.

Primary options may provide substantial benefit, but only in the right context. A primary option is not for everyone, but if you read the entry and find that you meet the criteria, consider adding the supplement to your stack.

Secondary options form another group of potentially beneficial supplements, but with less evidence for their effects. They could work or be a waste of money. Keep them in mind, but think twice before incorporating them into your stack.

Inadvisable supplements have either been shown to be ineffective, marketing claims notwithstanding, or are considered too risky. Do not add them to your stack. At best, they will be a waste of money; at worst, they can cause you harm.

Now that you have been presented with various supplements worthy of your interest, the time has come to combine them based on your objective. We will guide you in **assembling your stack**.

Then comes the **FAQ**, in which we cover common questions that may arise when assembling your stack.

Lastly, we include information on **precautions and troubleshooting**.

With all this combined, you should be able to identify and assemble the supplement stack best suited to your objective.

Core Supplements

Creatine

Why it's a core supplement

Creatine monohydrate can boast a lot of strong evidence for both its safety and its ability to increase power output and anaerobic endurance.

Supplementing creatine monohydrate increases the body's creatine stores, which are located primarily in the skeletal muscles. Your cells use creatine to regenerate adenosine triphosphate (ATP), life's energy currency, before they turn to burning glucose. In short, more creatine helps your muscles perform better under pressure (it helps you knock out those last few reps).

Creatine was also investigated for its effects on androgens, with most studies concluding that it had none. Moreover, the increase in testosterone reported by two studies (22% and 22.7%) is too small to have much of an effect on muscle gain or exercise performance. A lone study noted a 40–56% increase in dihydrotestosterone (DHT) with no change in testosterone — a puzzling result that has yet to be replicated.

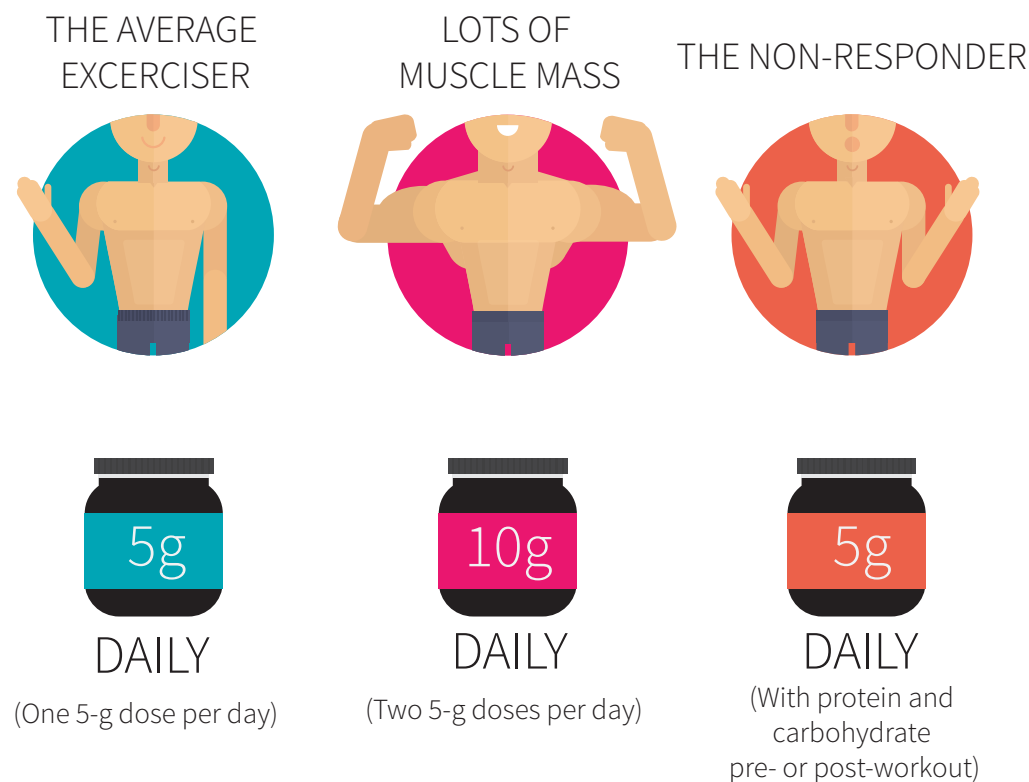
Decades of research vouch for creatine's safety. Its only potential side-effects are nausea, stomach cramps, and diarrhea from too large a dose. Creatine causes some water retention, but if you exercise, with time the proportion of added muscle to added water will increase.

How to take it

Take **creatine monohydrate**. Other forms of creatine may be more *expensive*, but studies have not found them to be more *effective*. Should you be particularly sensitive to creatine's digestive side-effects, which include nausea and cramping, try drinking more water. You can also split your daily dose and take it with meals. Finally, you could try *micronized* creatine monohydrate, which has for other advantage to be more soluble.

The standard daily dose is 5 g/day. This is enough to improve power output. People with more muscle mass may benefit from as much as 10 g/day, but this claim is not fully supported by the evidence. To supplement 10 g/day, take 5 g twice a day.

Figure 1: Creatine dose troubleshooting



Loading creatine means taking a high daily dose for a few days (e.g. 5 g five times a day for 5 days) before moving down to a smaller maintenance dose, which can be taken indefinitely. This is not necessary for effective supplementation, however; benefits may be felt sooner through loading, but they normalize after a few weeks.

Some people are creatine nonresponders: The creatine they ingest largely fails to reach their muscles. Alternate forms of creatine (such as creatine ethyl-ester) have been marketed as addressing this issue, but they lack scientific support. Currently, the best way to lessen creatine nonresponse is still to take creatine with a meal high in both protein and carbohydrate, preferably close to a time of muscle contraction (i.e. before or after your workout, if you work out).

Note that even if supplemental creatine fails to enter your muscles it can still benefit you in other ways, such as by improving your body's methylation status.

If you are not a creatine nonresponder, you need not worry about timing supplementation, though you should remember that taking your dose with a meal lowers the risk of an upset stomach. Creatine can be added to any liquid (even hot coffee or tea), but it must be drunk within the day, as creatine in liquids degrades into creatinine over time (the higher the temperature and the lower the pH, the faster the degradation). Do not buy liquid creatine products.

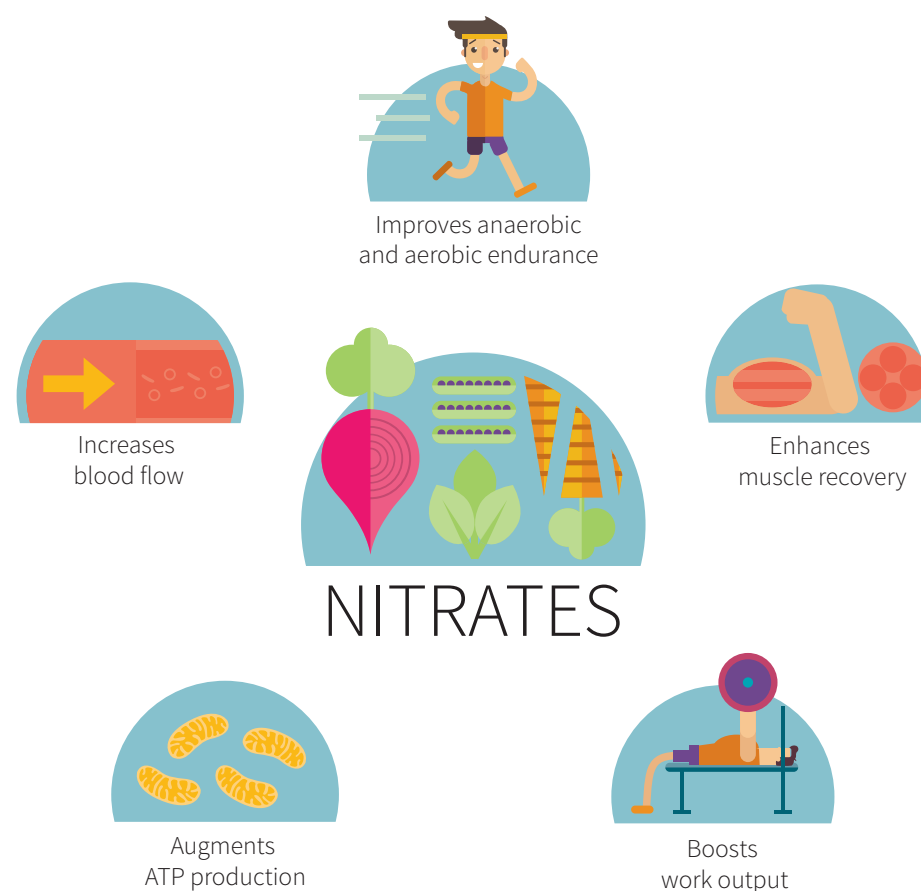
Nitrates

Why they're a core supplement

Nitrates can be found in different foods. Beetroot and leafy green vegetables are especially rich in nitrates. Nitrates break down into nitrites, which circulate in the body and are turned into nitric oxide (NO) as needed. Elevated NO levels during exercise provide a variety of benefits.

Nitrate supplementation has been shown to improve aerobic endurance, energy output, blood flow, and muscle recovery between bouts of exercise. It may also benefit anaerobic endurance, especially in beginners. Nitrates improve the body's ability to produce adenosine triphosphate (ATP) from food. [As aforementioned](#), ATP has been called life's energy currency: It powers your cells, including those composing your muscles.

Figure 2: Exercise-related benefits of nitrates



Nitrates do not exist as isolated dietary supplements, unfortunately, because of regulations against high quantities of sodium nitrate (a food additive frequently added to meat products). Instead, nitrate supplementation should take the form of a pre-workout meal incorporating leafy greens or beetroot.

Most studies on the performance-enhancing properties of nitrates used beetroot juice. Beetroot powder is also an option ($\frac{1}{8}$ the weight of raw beetroot), but not in capsules, as you would need to take too many. Cooking does not appear to reduce the bioavailability of nitrates from beetroot.

Leafy greens are often rich in [vitamin K](#), a fat-soluble vitamin that helps with blood clotting and so might decrease the effectiveness of blood thinners such as warfarin (Coumadin). If you take a blood thinner, you should consult with your doctor before consuming high levels of leafy greens.

Due to their goitrogen content, cruciferous vegetables can reduce thyroid hormone production if regularly consumed in high quantities, such as those needed for nitrate supplementation. If you eat a lot of cruciferous vegetables (such as cabbage, collard greens, or kale), make sure to also ingest enough [iodine](#) — through iodine-rich foods (such as cod, shrimp, milk, yogurt, or cottage cheese), iodine-fortified foods (such as iodized salt), or supplements (75–150 mcg/day).

How to take them

Nitrates are best supplemented through vegetables 1–2 hours before exercise. Since the bacteria in saliva play a role in activating dietary nitrates, do not use an antibacterial mouthwash too often, and especially not shortly before consuming nitrate-rich foods.

Vegetables sorted by nitrate content (mg per 100 g)

Nitrate content	Vegetables
Very high (250+)	Arugula/rocket, collard greens, dill, turnip greens
High (100 to <250)	Beetroot, celeriac, celery, Chinese cabbage, endive, fennel, kale, kohlrabi, leek, lettuce, mustard greens, parsley, spinach
Moderate (50 to <100)	Cabbage, savoy cabbage, turnip
Low (20 to <50)	Broccoli, carrot, cauliflower, cress, cucumber, chicory, pumpkin
Very low (<20)	Artichoke, asparagus, broad bean, eggplant, garlic, green bean, mushroom, onion, pea, pepper, potato, summer squash, sweet potato

Adapted from [Jones. Sports Med. 2014 May; 44 Suppl 1: S35–45](#) and [Hord et al. Am J Clin Nutr. 2009 Jul; 90\(1\): 1–10.](#)

Consuming these vegetables in liquid form (juice, shake, or purée) will increase the rate of nitrate absorption, since solid food particles take longer to digest. Aim for 6.4–12.8 mg of nitrates per kilogram of bodyweight (2.9–5.8 mg/lb). This corresponds to approximately:

- 440–870 mg for a 150-lb person
- 580–1,160 mg for a 200-lb person
- 730–1,450 mg for a 250-lb person

Nitrate-rich vegetables

Nitrate-rich vegetables	Content in mg per 100 g	
	Nitrates	Oxalate
Arugula/rocket	332.3	7.1
Dill	293.6	935
Turnip greens	284.5	50
Collard greens	254.5	450
Lettuce	189.3	330
Mustard greens	187.5	130
Radish	186.8	480
Beetroot	144.6	74.9
Chinese cabbage	138.8	1
Kale	137.5	20
Spinach	127.3	542.6
Celery	112.2	24.5

Adapted from [Lidder et al. *Br J Clin Pharmacol.* 2013 Mar](#), [Griesenbeck et al. *Nutr J.* 2009; 8](#), [Tamme et al. *Food Addit Contam.* 2006 Apr](#), and [Santamaria et al. *J. Sci Food Agric.* 1999 Oct](#).

Most vegetables rich in nitrates are also rich in oxalate, which can bind to minerals in the intestine and increase the risk of kidney stones. You need not ban all oxalate from your diet, but if you consume training-high amounts of nitrates more than twice a week, favor oxalate-poor vegetables. Also, keep in mind that cooking reduce oxalate content.

In any case, high amounts of oxalate-rich foods should be consumed away from meals (and from any mineral supplement you might be taking).

Vegetables sorted by oxalate content (mg per 100 g)

Oxalate content	Vegetables
Very high (100+)	Collard greens, dill, lettuce, mustard greens, radish, spinach, swiss chard
High (10 to <100)	Beetroot, cauliflower, celery, kale, turnip greens
Moderate (2 to <10)	Arugula/rocket, asparagus, carrot, cress
Low (<2)	Asparagus, cabbage, Chinese cabbage, radicchio, sweet potato

Because [glutathione](#) may slow down the rate of NO breakdown in the bloodstream, adding 200 mg of glutathione to your nitrates might prove synergistic.

Protein

Why it's a core supplement

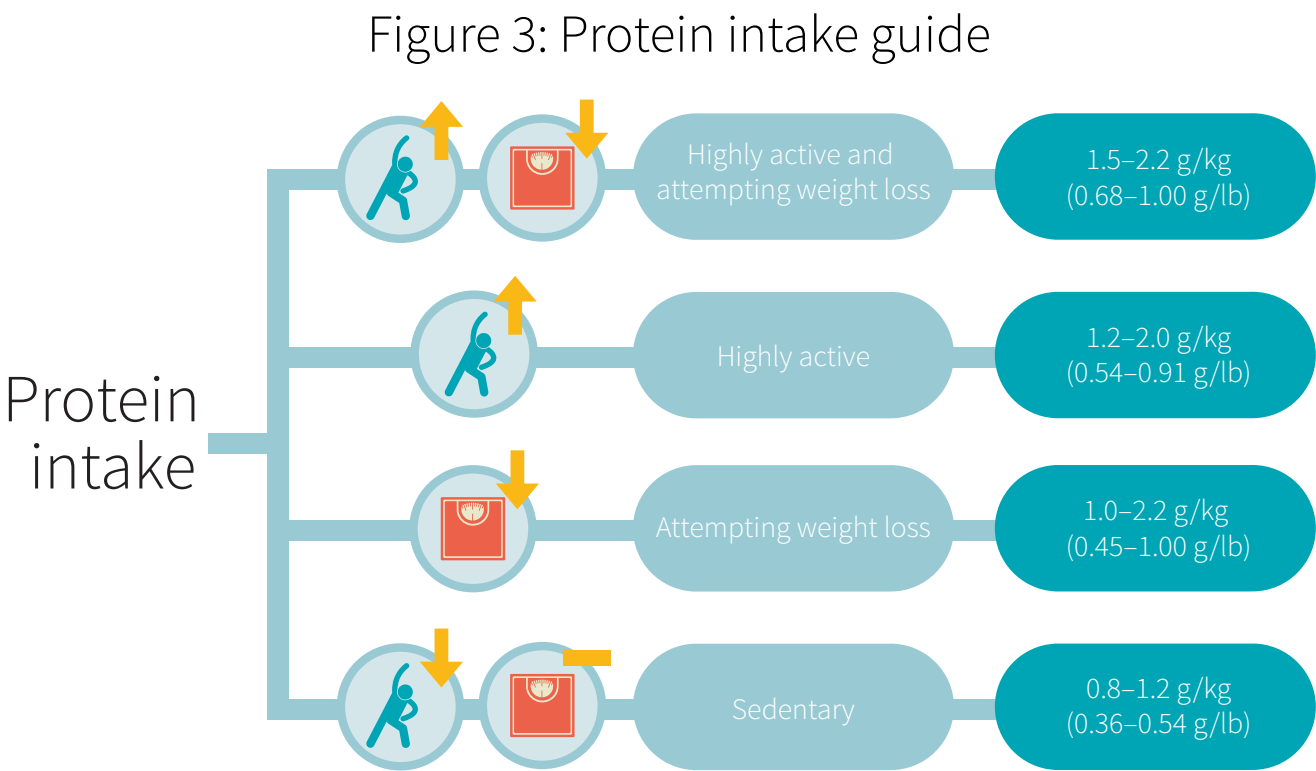
Any protein found in food or supplements is called dietary protein. The protein you ingest is digested into amino acids, which are then recombined as muscle fibers, among other things. Some amino acids (chiefly [leucine](#)) are especially important for muscle growth.

To maximize muscle growth and exercise performance, you need to consume enough protein. Whole food is best, but if your food intake does not cover your daily protein needs, you could add a supplement, preferably a powder.

[Whey](#) and [casein](#) powders are both derived from milk protein (which is 20% whey and 80% casein). If you are neither lactose intolerant nor vegan, look for a whey protein concentrate that is at least 80% protein. Whey is cheap and very anabolic (good for building muscle). Casein is more expensive, less anabolic, but more anti-catabolic (good for preserving muscle). Since casein digests slowly, it is often seen as the ideal protein to consume before sleep.

But what if you are lactose intolerant or vegan? Fortunately, you can still supplement protein powders. Whey isolates contain very little lactose. For vegans, two popular options are [soy](#), a complete protein, and a 70:30 pea:rice blend, which is seen as the vegan alternative to whey due to their similar amino acid profiles. Depending on processing techniques, the estrogenic isoflavone content can be greater in soy isolate than in soy concentrate, but it is still too small to elicit any significant hormonal response when as much as 200 g of soy protein isolate is consumed each day.

How to take it



- 1) If your BMI is more than 30, use your goal weight for calculations.
- 2) 0.8 g/kg may not be sufficient for elderly people losing muscle.
- 3) A higher protein intake does not seem to have negative health effects in people without kidney issues.

In the United States, the recommended daily allowance (RDA) for protein is 0.8 g per kilogram of bodyweight (0.36 g/lb/day). This amount will allow sedentary adults to maintain muscle mass without losing or gaining weight, but recent research suggests that people with fitness goals may benefit from more. Both 1.2–2.0 g/kg/day (0.54–0.91 g/lb/day) and 1.4–2.0 g/kg/day (0.64–0.91 g/lb/day) have been recommended.

Daily protein intake

Bodyweight	0.36 g/lb (0.8 g/kg)	0.54 g/lb (1.2 g/kg)	0.64 g/lb (1.4 g/kg)	0.91 g/lb (2.0 g/kg)
100 lb (45 kg)	36 g	54 g	64 g	91 g
150 lb (68 kg)	54 g	81 g	96 g	137 g
200 lb (91 kg)	72 g	108 g	128 g	182 g
250 lb (113 kg)	90 g	135 g	160 g	228 g

Adapted from *J Acad Nutr Diet*. 2016 Mar; 116(3): 501–28.

Unless you have a pre-existing condition that affects your liver or kidneys, the intakes in the above table will not harm those organs. However, if you have a BMI of more than 30, you may want to calculate your protein requirements based on your goal bodyweight rather than your current bodyweight, so as to avoid overeating.

If you work out in a fasted state, 20 g of protein within the 2 hours following your workout will help preserve muscle mass. Do not otherwise worry about timing, but spread your protein intake over a few meals so as to ease digestion and provide your body with a consistent flow of amino acids.

Sugars and Other Carbohydrates

Why they're a core supplement

Sugars are quickly digested carbohydrates. In addition to providing your muscles with readily usable energy, they cause a sharp rise in insulin, which improves blood flow and [the effects of nitric oxide](#) — two benefits that contribute to peak physical performance.

Sugars are especially beneficial to anaerobic exercise, such as weightlifting or sprints, but they can also fuel the longer endurance events, such as marathons. Workouts of mild intensity and duration, however, can simply be fueled by other carbohydrates consumed throughout the day.

How to take them

You can use the chart below to help set your total daily carbohydrate intake (in grams) according to your bodyweight (in kilograms or in pounds).

Daily carbohydrate intake

Intensity	Activity	g/kg	g/lb
Low	Skill-based or general activities	3–5	1.4–2.3
Moderate	Exercise program (1 hr)	5–7	2.3–3.2
High	Endurance program (1–3 hr)	6–10	2.7–4.5
Very high	Extreme commitment (4–5+ hr)	8–12+	3.6–5.4+
Strength athletes	Resistance training (weightlifting, bodybuilding, etc.)	4–7	1.8–3.2

Adapted from [J Acad Nutr Diet. 2016 Mar; 116\(3\): 501–28.](#)

Athletes can use the table below to optimize the timing of their carbohydrate intake. Other people can just remember the following sugar protocol: If you exercise for 45 minutes to 2 hours, aim for 30–60 g/hr; if you exercise longer, aim for 60–90 g/hr.

Carbohydrate intake timing

	Situation	Time	Carbohydrates
Before	Loading	36–48 hr before the event	4.5–5.4 g/lb/day (10–12 g/kg/day)
	Pre-event	1–4 hr before the event	0.5–1.8 g/lb (1–4 g/kg)
During	Brief exercise	≤45 min	Unnecessary
	Sustained high-intensity exercise	45–75 min	Sugars: ≤30 g/hr
	Endurance exercise, including stop-and-start sports such as basketball, soccer, etc.	60–150 min	Sugars: 30–60 g/hr
	Ultra-endurance exercise	>150 min	Sugars: ≤90 g/hr
After	Speedy refueling	≤10 hr between two fuel-demanding sessions	0.5 g/lb/hr (1.0–1.2 g/kg/hr) for the first 4 hours then resume normal intake
	Typical refueling	>10 hr between two fuel-demanding sessions	0.5–0.7 g/lb (1.0–1.5 g/kg) over the first 30 minutes then resume normal intake

Adapted from *J Acad Nutr Diet*. 2016 Mar; 116(3): 501–28.

Athletes should be careful not to overdo their sugar consumption before a game or competition. Too much at once may cause temporary reactive hypoglycemia, a short period of low blood sugar, which could hurt sports performance. If this is something you regularly experience, switching to a pre-game carb source with a low [glycemic index](#) value (GI) may provide you with a more even source of energy.

As a general rule, eating a high-carb meal before exercise decreases the need for a sugary drink during exercise. Conversely, the less carbohydrate in your regular diet, the greater the performance-enhancing effects of a sugary drink during exercise.

During exercise, drinking a mix of two or more sugars or other simple carbohydrates (glucose, fructose, maltose, etc.) may help increase the rate at which they are absorbed and oxidized for energy. Although maltodextrin is not a simple sugar, its absorption rate is similar to that of dextrose (D-glucose), so it can be used as part of your mix.

Your exercise beverage should be 6–8% carbohydrate (a lot less than most sports drinks) to optimize absorption and oxidation and minimize the risk of an upset stomach. Also to avoid stomach discomfort, no more than half your mix should be fructose, and you should steer clear from sodas, since their carbonation and acidity can cause gastrointestinal upset during exercise.

The main factors to influence muscle glycogen resynthesis are: carbohydrate type, ingestion rate, intake timing, coingestion of carbohydrate and [protein](#), and caffeine intake. Ingesting a high-GI food or beverage at a rate of 1.0–1.2 g/kg/hr (0.5 g/lb/hr) for the first 4 hours following exercise will help maximize the speed of glycogen resynthesis. Adding protein (at a CHO:PRO ratio of 3:1 or 4:1) and/or caffeine may slightly increase the speed of glycogen resynthesis, too.

The speed of glycogen resynthesis matters mostly for people who exercise more than once a day. As a rule, carbohydrate supplementation during and after exercise is more of a concern for endurance athletes than for strength athletes; the latter should focus on consuming adequate carbohydrate throughout the day and before exercise. For both types of athlete, the shorter the rest between workouts, the more important intake timing becomes.

Primary Options

Beta-Alanine

Why it's a primary option

When ingested, the nonessential amino acid beta-alanine binds with the essential amino acid histidine to create carnosine. Carnosine has anti-aging and antioxidant properties; it also buffers lactic acid during exercise, which delays muscle fatigue. Essentially, beta-alanine supplementation can improve endurance.

But there's a catch. Beta-alanine only helps with intense exercise, in which lactic acid kicks in after about a minute. Imagine running a fast lap around the track or doing a volume set of squats that feels like the longest sixty seconds of your life. Beta-alanine supplementation will mostly benefit athletes who exercise hard in the 1–4-minute range.

How to take it

Take 4–6 g/day.

If you have a long workout planned, aim for the higher end of that range. It is better to take this supplement even on rest days, but skipping one or two days a week is not a major issue. While beta-alanine can be taken at any time of the day, it may be better absorbed with a meal.

A common side-effect of beta-alanine is paresthesia (a tingling sensation, as when your leg “falls asleep”). Taking smaller doses throughout the day or using time-release formulations can help reduce the prickling sensation on the skin (especially the face). Paresthesia is harmless, if unpleasant.

Caffeine with Theanine

Why it's a primary option

In addition to staving off sleep through its antagonist action on adenosine receptors, caffeine can increase muscular power output by stimulating

production of adrenaline and dopamine — two chemicals responsible for the euphoric feeling you remember from your first-ever cup of coffee.

Unfortunately, with frequent use, caffeine loses most of its power-enhancing effects, though it can still help stave off sleep. Infrequent caffeine intake is key to experiencing the stimulatory effect every time. This is why caffeine is not recommended as a core supplement.

Even though coffee is a popular beverage worldwide, caffeine is not innocuous. [Regular consumption leads to tolerance and often to dependence and withdrawal.](#) Moreover, caffeine interacts dangerously with monoamine oxidase inhibitors (MAOIs), a kind of antidepressant, and [several other pharmaceuticals](#), notably tizanidine.

Caffeine can also decrease blood lithium levels. Suddenly eliminating all caffeine from your diet can cause lithium levels to rise. [If you are on lithium, keep your day-to-day caffeine intake roughly the same;](#) if you wish to stop taking caffeine, talk with your doctor about slowly weaning yourself from it.

Theanine (the amino acid L-theanine) can reduce the overexcitability associated with caffeine without decreasing caffeine's stimulatory effect. In fact, the improvements in concentration (focus and attention span) induced by caffeine and theanine respectively have been shown to be synergistic. Therefore, while theanine on its own does little to enhance exercise performance, it becomes a primary option when combined with caffeine.

How to take it

Caffeine is most effective when supplemented by people who seldom take any. If you get excited after a cup of coffee, you'll probably benefit from caffeine's effects at the gym.

No more than twice a week, take [caffeine](#) (400–600 mg) with [theanine](#) (300 mg) half an hour before an especially strenuous workout. If you begin to develop a caffeine tolerance, drop down to one dose a week. It may be necessary to stop all caffeine use for at least a month to regain caffeine sensitivity.

Supplementing caffeine on an empty stomach can increase the rate of absorption, but it can also cause gastrointestinal upset. With its half-life of 5 to 6 hours, caffeine can disrupt sleep when consumed in the evening. Even if it does not prevent you from falling asleep, caffeine will impair the *quality* of your sleep.

Secondary Options

Ashwagandha

Why it's a secondary option

Adaptogens are supplements that can reduce the mental and physical effects of stress, including fatigue, depression, and anxiety. The most popular and well-researched adaptogens are ashwagandha (*Withania somnifera*), Panax ginseng, and Rhodiola rosea. These herbs have been the subjects of many studies, but rarely in the context of increased muscular power and never as pre-workout supplements. Preliminary evidence suggests that they can lower the perception of fatigue when taken before exercise, but more research is needed to confirm this effect.

The ashwagandha extract most used in studies is called KSM-66. It is a water-based extract standardized to 5% withanolides. Supplementation of 300 mg twice a day (i.e. 600 mg/day) has been shown to bolster muscle gains and reduce blood cortisol levels.

How to take it

To supplement **ashwagandha**, take 300 mg of the KSM-66 extract twice a day (i.e. 600 mg/day), including one dose 30–60 minutes before exercise. Alternatively, take your whole 600-mg daily dose 30–60 minutes before exercise. Should you purchase another extract standardized for withanolide content, aim for 30 mg of withanolides per day. If you only have access to the root powder, try 5 g/day.

To supplement **Panax ginseng**, take 200–400 mg of an extract containing 1–3% ginsenosides 30–45 minutes before exercise.

To supplement **Rhodiola rosea**, take 80–160 mg of SHR-5 (an extract standardized to 3% rosavins and 1% salidroside) 30–45 minutes before exercise. Up to 500 mg can be taken in anticipation of a particularly stressful event.

BCAAs

Why they're a secondary option

Branched-chain amino acids (BCAAs) are three of the nine amino acids essential to humans. Since you ingest BCAAs each time you ingest [protein](#), supplementing BCAAs in isolation is mostly redundant. For example, 100 g of a whey protein concentrate can contain 11 g of [leucine](#), 6 g of [isoleucine](#), and 6 g of [valine](#), so 23 g of BCAAs (the numbers vary between supplements).

Many studies have investigated the effects of BCAA supplementation on exercise, and these effects proved minimal. Mostly, BCAAs were found to relieve cognitive fatigue during exercise lasting more than two hours. This could be useful for athletes who need to maintain hand-eye coordination over a long game (hockey or football players, for instance). In novices, the reduction in mental fatigue might also translate into an increase in physical endurance.

On the whole, BCAAs have only two potential advantages over protein powders. **First**, they are less likely to cause cramping or nausea when consumed before exercise. **Second**, they are free of tryptophan (an amino acid that might promote exercise-related fatigue).

How to take them

To supplement BCAAs, take 10–20 g before exercise, in water or a sugary drink. A ratio of 2:1:1 (leucine:isoleucine:valine) is often recommended based on two studies that actually used a 2.3:1:1.2 ratio. Both studies compared this 2.3:1:1.2 mix to a placebo (dextrin); different ratios were not compared, so the ideal ratio is unknown.

Cholinergics

Why they're a secondary option

Acetylcholine is a major neurotransmitter involved in memory formation and muscle contractions. A supplement is said to be cholinergic when it increases the activity of acetylcholine, especially in the brain. Elevated acetylcholine levels in neurons are associated with stronger muscle contractions, but cholinergics themselves are not well-researched in this context. There is currently no strong evidence that cholinergics increase muscular power output.

[CDP-choline](#) and [alpha-GPC](#) can provide the brain with the choline it needs to produce more acetylcholine. There is some preliminary evidence that alpha-GPC can increase power output, but this result has yet to be replicated. As for the much cheaper [choline bitartrate](#), little of it reaches the brain, so little of it gets converted into acetylcholine.

[Huperzine-A](#) can inhibit acetylcholinesterase, an enzyme that breaks down acetylcholine. Its half-life exceeds 24 hours, so it accumulates in the body when taken daily, which is problematic because of a dearth of long-term studies. The risk exists that, with time, the body could adapt by producing more acetylcholinesterase, which would lead to reduced acetylcholine levels and a withdrawal period after huperzine-A supplementation has ceased. While the doses used in the studies (0.2–0.8 mg) were deemed safe in the short term, long-term supplementation cannot be recommended.

How to take them

To supplement **alpha-GPC**, take 300–600 mg 30–45 minutes before exercise.

To supplement **CDP-choline**, take 250–500 mg 30–45 minutes before exercise.

HMB

Why it's a secondary option

Beta-hydroxy beta-methylbutyrate (HMB) is a product of the amino acid [leucine](#) thought to be anti-catabolic (to help prevent muscle breakdown), mostly during periods of caloric restriction, inadequate protein intake, or heavy exercise. For that purpose, HMB might be twenty times as potent as leucine (gram for gram).

By reducing muscle damage, HMB may allow for faster recovery times both during and between workouts, leading to faster increases in performance. People just starting to exercise and people undertaking high volumes of work are most likely to benefit from HMB. It is uncertain, however, whether HMB has any effect in people who consume enough leucine-rich protein.

Calcium HMB is the most common form of HMB. While another form, known as HMB free acid (HMB-FA), has shown some promise in a few recent

studies, the drastic difference in potency between the two forms — from inconsistent benefits for calcium HMB to near steroid-like effects for HMB-FA — needs to be confirmed by additional research.

How to take it

Take 3 g of HMB (calcium HMB or HMB-FA) per day. On workout days, take the whole daily dose 1 hour before exercise.

Note that it may take you up to 2 weeks to feel the full effects of HMB, particularly if you are engaging in high-intensity workouts.

Nitric Oxide Boosters

Why they're a secondary option

Elevated nitric oxide (NO) levels are associated with improved blood flow, muscle growth, and more efficient energy production. NO boosters form a category of supplements meant to increase NO levels in the body. Some common NO boosters are [arginine](#), [citrulline](#), and [agmatine](#). [Nitrates](#) from beetroot and leafy green vegetables are not included in this category.

Unfortunately, arginine, citrulline, and agmatine all have issues. Absorption of arginine by the intestines is limited, and much is eliminated from the body before it can reach the muscles. Citrulline is better absorbed and is converted into arginine in the kidneys, but in a controlled manner (rather than a huge spike in NO levels, you can expect a steady increase of lesser magnitude). Agmatine is a newer, promising supplement, but one that lacks human evidence for its effects.

Today's NO boosters have been marketed based on promising initial evidence, but their effects are unreliable at best. They should only be considered for supplementation if dietary nitrates are not an option. Should it be the case, citrulline is the best alternative among bad ones.

Note: NO boosters should not be taken with [yohimbine](#), as they counteract the effects of this fat burner (unlike [nitrates](#), which increase NO through another pathway).

How to take them

Half an hour before exercise, take one of these:

0.5–1.5 g of **agmatine**.

6 g of **arginine**. Arginine supplementation can cause diarrhea, so start with 3 g before working your way up. Concomitant consumption of caffeine increases the risk of diarrhea.

6 g of **citrulline** (or 10 g of citrulline malate). Thanks to better absorption rates, intestinal side-effects are less likely than with arginine.

Because [glutathione](#) may slow down the rate of NO breakdown in the bloodstream, adding 200 mg of glutathione to your NO booster might prove synergistic.

Inadvisable Supplements

Glutamine

Glutamine is an amino acid that plays an important role in muscle cells. In fact, *in vitro* studies (studies done in a test tube or a petri dish) require glutamine to keep cells alive. When glutamine is added to muscle cells *in vitro*, protein synthesis increases.

The problem is getting glutamine to muscle cells in the first place. After ingestion, glutamine does not reach the muscle directly. Instead, it is taken up by the intestines, then transported to the liver, which releases it to other tissues in the body on an as-needed basis.

Therefore, while glutamine supplementation will benefit the digestive system and the liver, it will not cause an increase in muscle protein synthesis and thus cannot be recommended for inclusion in a performance-enhancing stack.

Sodium Bicarbonate

Sodium bicarbonate, also known as baking soda, can improve muscular endurance. Alas, it can also cause severe stomach upset if supplemented improperly. Keep in mind that one dose is 300–400% of the daily recommended sodium intake. Sodium bicarbonate and [beta-alanine](#) have similar effects, but taking both at once will not be more effective than taking beta-alanine by itself.

Due to the relatively high potential for side-effects, sodium bicarbonate cannot be recommended as a primary or secondary option.

Taurine

The amino acid L-taurine gave its name to Red Bull (*taurus* is Latin for “bull”) and can also be found in other energy drinks. There is precious little evidence, however,

to support its use to enhance exercise performance, except maybe in people with heart failure. Otherwise, the most you *might* gain from taurine supplementation is a reduction in the muscle cramping associated with fat burners.

If you choose to supplement taurine, do not take more than 3 g/day.

Testosterone Boosters

There is precious little human evidence to support the efficacy of testosterone boosters. Studies are seldom replicated, and when replicated seldom draw the same conclusions. Furthermore, even if a supplement can coax your body into producing more testosterone, it can only do so within your physiological limits — do not expect steroid-like effects.

Some supplements claiming to boost testosterone — including [maca](#), [fenugreek](#), and [Tribulus terrestris](#) — actually enhance libido. This effect can translate into increased confidence, maybe directly (increased libido = increased confidence), more probably indirectly (increased libido = a feeling that the supplement is working = increased confidence).

The positive influence of confidence on exercise performance has been noted in several dedicated studies, so if libido boosters improve your power output due to improved energy and mood, they may have a role to play in your stack.

Assembling Your Stack

Incorporating Core Supplements

The core supplements in this stack are [creatine monohydrate](#), [dietary nitrates](#), [protein](#), and [carbohydrates](#).

1–2 hours before exercise, take creatine monohydrate (5 g) and dietary nitrates (6.4–12.8 mg per kilogram of bodyweight, so 2.9–5.8 mg/lb). Consume protein throughout the day (1.2–2.0 g per kilogram of bodyweight, so 0.54–0.91 g/lb/day), with the help of protein powders if necessary. Consult the [Sugars and Other Carbohydrates](#) section to set your carbohydrate intake.

The core supplements are recommended for most people; their efficacy and safety are backed by a significant body of evidence. Take them for a couple of weeks before you consider making any modification, such as adding one of the following options.

Incorporating Options

For weightlifters (muscle growth and power)

In addition to the core supplements, take [caffeine](#) (400–600 mg) with [theanine](#) (300 mg) half an hour before an especially strenuous workout, no more than twice a week. After one month, consider adding either [alpha-GPC](#) (300–600 mg) or [CDP-choline](#) (250–500 mg). Optionally, either of these two cholinergics can be taken half an hour before training every day.

For athletes who exercise for one hour or more (alertness and endurance)

In addition to the core supplements, take [BCAAs](#) (10–20 g) before exercise. You can also take [beta-alanine](#) (5 g/day) with a meal.

For people undergoing intense training, such as HIIT

In addition to the core supplements, take [beta-alanine](#) (5 g/day) with a meal and [ashwagandha](#) (600 mg of the KSM-66 extract) about one hour before

exercise. Soon after exercise, eating or drinking a mix of protein (0.15–0.25 g/kg, so 0.07–0.11 g/lb) and carbohydrate at a CHO:PRO ratio of 3:1 or 4:1 will help replenish glycogen stores and promote muscle recovery.

Other options

If you work out in a fasted state, 20 g of [protein](#) within the two hours following your workout will help preserve muscle mass. If you fail to get enough protein during the day, [HMB](#) (3 g/day) or [BCAAs](#) (10–20 g/day) can also help you preserve muscle mass.

People with high levels of muscle mass can increase the creatine dose to 10 g.

Should you find it impractical to consume 500 g of nitrate-rich vegetables, you could try 6 g of [citrulline](#) (or 10 g of citrulline malate) instead.

Because [glutathione](#) may slow down the rate of NO breakdown in the bloodstream, adding 200 mg of glutathione to your nitrates or citrulline might prove synergistic.

[Adaptogens](#) have not been shown to negatively interact with any compound listed in this guide. They can be added to any stack if mental fatigue is a problem during or after a workout.

FAQ

Can I add to my stack a supplement not covered in this guide?

Supplement your current stack for a few weeks before attempting any change. Talk to your doctor and [research each potential new addition](#) in advance. Check for known negative interactions with other supplements in your current stack, but also for synergies. If two supplements are synergistic or additive in their effects, you might want to use lower doses for each.

Can I modify the recommended doses?

If a supplement has a recommended dosage range, stay within that range. If a supplement has a precise recommended dose, stay within 10% of that dose. Taking more than recommended could be counterproductive or even dangerous.

Should I take my supplements with or without food? In the daytime or the evening?

Certain supplements have strong evidence for how and when to take them, which we have noted where applicable. For most supplements, however, this research is either mixed or absent. It is always a good idea to start with a low dose of a supplement, so as to minimize the harm of taking it during the day (e.g. tiredness) or in the evening (e.g. insomnia).

Keep in mind that taking a supplement with food does not mean that all foods will have the same effect on its absorption, digestion, or metabolism. The most basic example is fat-soluble vitamins (A, D, E, and K), which will absorb better with a small meal containing fat than a large meal containing little to no fat.

I have an iron stomach. I have never felt nauseous from supplements. Do I still need to take precautions to avoid gastrointestinal upset?

If you have never had any issues with nausea or vomiting, you may have an easier time ingesting large doses of certain supplements. Nevertheless, it is not a good idea to disregard the warnings on a product.

I thought creatine and caffeine negated each other, so why are they in the same stack?

Although not all studies agree, a high dose of caffeine (5 mg per kilogram of bodyweight, so about 2.3 mg/lb) might partially negate the benefits of creatine, but only during a [creatine loading phase](#) and when both supplements are taken at the same time. This potential issue can be sidestepped by consuming your creatine and caffeine several hours apart or by skipping the optional creatine loading phase.

Can I take caffeine without theanine, or vice versa?

Caffeine is an effective stimulant on its own. Adding theanine will enhance your focus. Theanine by itself provides no exercise-related benefit.

Why do you have entries for BCAAs and HMB but not leucine?

[BCAAs](#) might alleviate cognitive fatigue when taken before a game, so they have a niche to fill. There is some evidence that [HMB](#) could be as anti-catabolic as twenty times the same amount of [leucine](#) when calories are restricted, in which case it would also have a niche to fill. With regard to anabolism, increasing your protein intake is more likely to help than leucine alone, so leucine doesn't really have a niche to fill.

Since the body makes carnosine out of beta-alanine and histidine, should I also supplement histidine?

It isn't necessary. If you consume enough protein, your muscles already have all the histidine they need to produce more carnosine.

Still, why beta-alanine? Wouldn't it be simpler to supplement carnosine directly?

Since carnosine simply gets digested into beta-alanine and histidine, and since your muscles already have enough histidine, carnosine supplementation has no advantage over beta-alanine supplementation — especially since beta-alanine is cheaper.

Precautions and Troubleshooting

Stack components are seldom studied together. The safest way to add supplements to your daily routine is one at a time, at least a couple of weeks apart, to better assess the effects (and side effects) of each new addition. Start at half the regular dose for a week, then slowly increase to the regular dose if you are not experiencing the desired effects.

Be aware that core-intensive exercises, supplementing on an empty stomach, dietary protein, and too much carbohydrate (especially fructose) can all increase the risk of gastrointestinal upset. Any stack that prevents you from maximally exerting yourself because of nausea or stomach pain is counterproductive.

To reduce the risk of gastrointestinal upset, take your supplements with enough water and (if possible) a small meal with a little fat and a higher-than-average carbohydrate content. You could also let more time pass between the moment when you take your supplements and the beginning of your workout.

Athletes should **never** take a supplement for the first time before a game or competition. Always try the supplement during practice first. Taking a seat on the bench to calm your stomach is much easier when you don't have to worry about winning a game.